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I am conducting a literature review on infrastructure-less emergency communication systems using LoRa or similar sub-GHz technologies. Please identify peer-reviewed papers that discuss decentralized, peer-to-peer, or mesh-based emergency communication architectures, including multi-hop routing, latency evaluation, reliability analysis, and use cases in disaster recovery, vehicular safety, or remote

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About Results

IEEE AI Overview

Several studies have explored infrastructure-less emergency communication systems using LoRa or similar sub-GHz technologies, focusing on decentralized, peer-to-peer, or mesh-based architectures. A LoRa-based mesh network has been proposed for off-grid emergency communications, particularly in disaster-stricken areas where traditional telecommunication infrastructures become unusable [1]. This system utilizes a modified ad hoc on-demand distance vector (AODV) routing protocol, enabling mesh networking in LoRa, and demonstrates its feasibility through text message transmission between mobile phones.

In a similar vein, the ClusterDuck Protocol, a new networking protocol and embedded software implementation, has been introduced for long-range, low-power mesh network infrastructure [2]. Although the protocol shows promise, its implementation has been found to have high error rates and bugs, leading to increased processing…

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References:

1. LoRa-based Mesh Network for Off-grid Emergency Communications
2. Empirical Evaluation of a LoRa Mesh Network for Emergency Communication Systems
3. Demo: Chat based Emergency Service via Long Range Wireless Communication (LoRa)
4. LOCATE: A LoRa-based mObile emergenCy mAnagement sysTEm
5. R-AODV: Reverse Ad-Hoc On-Demand Distance Vector Routing Algorithm With LoRa Mesh Based for Emergency Alert System
6. UAV-Aided Emergency Environmental Monitoring in Infrastructure-Less Areas: LoRa Mesh Networking Approach

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Recommended Result

LoRa-based Mesh Network for Off-grid Emergency Communications

Khazmir Camille Valerie G. Macaraeg; Calvin Artemies G. Hilario; Charleston Dale C. Ambatali **All Authors**

Publisher: IEEE | 2020 | Conference Paper

AI Summary

- Philippines prone to natural disasters, disrupting telecom networks.
- Proposes LoRa-based mesh network for emergency communication.
- Modified AODV routing protocol uses RSSI as metric, evaluated by PDR.

Abstract

The Philippines, being one of the most disaster-stricken countries in the world, is susceptible to property and infrastructure damage caused by natural calamities. During these times, most of the major telecommunication network infrastructures in the affected area become unusable. Thus, developing a robust, low-power, decentralized emergency communication network becomes necessary in order to coordinate the emergency response and relief efforts. In this paper, we propose a mesh network based on LoRa technology as a viable emergency communication tool. We implement a modified ad hoc on-demand distance vector (AODV) routing protocol to enable mesh networking in LoRa. The modified AODV uses the received signal strength indicator (RSSI) as the routing metric. The performance of the proposed system was evaluated based on the packet delivery ratio (PDR). We demonstrate the feasibility of the developed system by transmitting text messages from one mobile phone to another through LoRa-based mesh network. Show Less

Recommended Result

Empirical Evaluation of a LoRa Mesh Network for Emergency Communication Systems

Brenton Poke; Suleyman Uludag **All Authors**

Publisher: IEEE | 2023 | Conference Paper

AI Summary

- Proposes ClusterDuck Protocol for disaster communications, a long-range, low-power mesh network infrastructure that's cost-effective.
- Current implementation has high error rates and bugs, but shows promising power consumption.
- Suggests improvements and transmission strategies to increase protocol effectiveness.

Abstract

Natural and man-made disasters are becoming more prevalent and increasing in danger as climate change continues unabated and resources get scarce. Whether it be flood or terrorism, disasters displace people and it remains difficult to reach those in need when the disaster does strike. For years, mobile phone networks have been integral in responding to emergencies, but they rely on costly infrastructure that is prone to outages or outright attack. We propose a long-range, low-power mesh network infrastructure based on a new networking protocol and embedded software implementation called the ClusterDuck Protocol, that seeks to fill the need of disaster communications telemetry gathering for a fraction of the cost of mobile phone networking infrastructure such as 4G LTE or 5G. While the protocol is simple, it was found to have an inadequate implementation at this stage. Error rates were high and bugs were found in implementation that led to an abnormal amount of time spent processing corrupted data, though power consumption was encouraging in spite of these conditions. We examine the strengths and weaknesses of this new protocol and suggest improvements and transmission strategies to be more successful. Show Less

Recommended Result

Demo: Chat based Emergency Service via Long Range Wireless Communication (LoRa)

Alisa Stiballe; Simon Welzel; Johannes Dorfschmidt; Darvin Schluter; Dominik Delgado Steuter; Jannik Lukas Hense; Florian Klingler **All Authors**

Publisher: IEEE | 2023 | Conference Paper

AI Summary

- Novel networking architecture for emergency communication in crisis situations.
- Ad hoc and distributed multi-hop networking approach keeps communication overhead low.
- Demo features LoRa long-range communication, web-based chat client, and novel data processing.

Abstract

Internet and mobile communication are the foundation of today's worldwide interaction and connectivity. Recently, several approaches have been introduced in order to add resilience to communication networks mainly by using low cost and commodity hardware by e.g., facilitating smart phones interconnected via LoRa communication. We extend those ideas by introducing a novel and versatile networking architecture for emergency communication in the context of crisis situations (e.g., when cellular coverage is gone) for mobile devices to keep communication overhead low by following an ad hoc and distributed multi-hop networking approach. In this work, we present a demo of our network to handle emergency communication. It features the long-range, wide area communication technique (LoRa), a web-based chat client with a user-friendly interface, a novel data processing approach, as well as a routing algorithm to ensure fast and efficient communication. Show Less

LOCATE: A LoRa-based mObile emergenCy mAnagement sysTEm

Luca Sciullo; Federico Fossemo; Angelo Trotta; Marco Di Felice **All Authors**

Publisher: IEEE | 2018 | Conference Paper

AI Summary

- Proposes Emergency Communication System (ECS) for infrastructure-less phone-based networks using LoRa technology.
- ECS includes mobile app for conveying minimal emergency data and dissemination protocol for spreading requests over multi-hop LoRa links.
- Evaluates performance through experimental study on LoRa technology's distance capabilities and simulation study comparing to state-of-the-art flooding schemes.

Abstract

During the occurrence of an emergency, being it the consequence of a natural disaster or of human activities, the pervasiveness of user-owned mobile devices (e.g. smartphones, tablets) turns into a precious help to convey data and services to all the people involved. As a result, several emergency-related mobile applications have been proposed on the market; however, they are often limited by the networking capabilities of the devices, since they are often based on short-range Device-to-Device (D2D) communication technologies (e.g. the Wi-Fi Direct), or on the cellular infrastructure, which might be not available on the emergency scenario. In this paper, we attempt to overcome both such issues, by proposing a novel Emergency Communication System (ECS) which operates over infrastructure-less phone-based networks, and guarantees long-range D2D communication thanks to the LoRa technology. The system, named LOCATE, includes a mobile application, through which users can convey minimal yet vital emergency-related data, and a dissemination protocol, spreading the emergency requests over multi-hop LoRA links. The performance of LOCATE have been evaluated through: (i) an experimental study, assessing the capability of LoRa technology to convey short, emergency messages over long distances, and a (ii) simulation study, demonstrating the effectiveness of the dissemination protocol on large-scale scenarios when compared to state-of-the-art flooding schemes. Show Less

R-AODV: Reverse Ad-Hoc On-Demand Distance Vector Routing Algorithm With LoRa Mesh Based for Emergency Alert System

Jirawat Thaenthong; Taiwit Pimsen; Kemmathad Chidchuea **All Authors**

Publisher: IEEE | 2023 | Conference Paper

AI Summary

- LoRa mesh network addresses limitations of traditional point-to-point systems.
- Mesh network employs R-AODV routing to adapt to node failures & route changes.
- System improves coverage & reliability for emergency communication, with average delay around 1s.

Abstract

This research introduces a LoRa mesh network for emergency notification systems, which addresses the challenges of the conventional point-to-point LoRa system, such as limited transmission range and lack of backup paths. The mesh network comprises nodes that serve as relays, facilitating multi-hop data transmission to the gateway. The network also adapts to node failures and route changes by employing the Reverse Ad-hoc On-demand Distance Vector (R-AODV) routing algorithm. The performance of the proposed system was assessed through three types of tests: point-to-point, multi-hop, and node failure. The results indicate that the LoRa mesh network improves network coverage and reliability for emergency communication. The average delay for sending alert messages from the gateway neighbor node to the gateway is 0.82s, from the alarm node to the gateway is 1.54s in normal conditions, and 11.65s in case of a gateway neighbor node failure. The user satisfaction score is 4.31 out of 5. Show Less

UAV-Aided Emergency Environmental Monitoring in Infrastructure-Less Areas: LoRa Mesh Networking Approach

Miaoxin Pan; Chongcheng Chen; Xiaojun Yin; Zhengrui Huang **All Authors**

IEEE Internet of Things Journal
Publisher: IEEE | 2022 | Journal

AI Summary

- UAV-aided system uses LoRa mesh networking for rapid environmental monitoring in disaster areas.
- Custom slotted ALOHA medium access mechanism enables efficient data transmission at 0.293 kb/s.
- System achieves line-of-sight transmission distance >10 km, delay <6 s, and meets real-time emergency response requirements.

Abstract

Rapid acquisition of environmental parameters of emergency sites is crucial for emergency response. However, in areas where public ground networks do not provide sufficient coverage or are destroyed by disasters, the rapid backhaul of monitoring data from emergency sites is difficult. Due to the urgent demand for data collection in such areas, we propose an unmanned aerial vehicle (UAV)-aided emergency environmental monitoring system using a LoRa mesh networking approach, which can quickly build a large range of wireless communication networks suitable for small data transmission. First, we designed a LoRa-based mesh network protocol by using a custom slotted ALOHA medium access mechanism. Second, we designed and implemented a custom software and hardware platform of the monitoring system. The system adopts a communication mode, including multiple sensor nodes (SNs), multiple relay nodes (RNs), a gateway (GW), and a network server. Finally, we demonstrated the performance of the protocol and the system by in-lab and outdoor tests. The results show that when data rate is 0.293 kb/s, the line-of-sight 1-hop transmission distance is more than 10 km, and the 1-hop transmission delay is less than 6 s; the end-to-end delay has an upper bound, which can meet the real-time requirement of emergency response; the least upper bound of throughput per slot of our mesh protocol is much higher than that of LoRaWAN. Evaluation results also demonstrate the network robustness to environmental changes. Show Less

Humanity Lifeline: A Resilient Communication and Sensor Network Framework for Disaster Response

Ozgun Pinarer; Omer Komili **All Authors**

IEEE Access

Publisher: IEEE | 2025 | Journal

AI Summary

- A hybrid communication system integrates BLE with LoRa and LoRaWAN to ensure robust and scalable networks under disaster conditions.
- The proposed LifeLinks system uses custom hardware with Renesas RA4W1 microcontrollers and LoRa-Router-Gateways for peer-to-peer and cloud connectivity.
- Experimental results show 99% successful message delivery, minimal data loss, and strong network reliability in high-traffic scenarios and disrupted environments.

Abstract

In the domain of disaster communication, the failure of conventional infrastructures such as cellular towers and power lines poses significant challenges to timely information exchange. Existing solutions often fall short in providing robust and scalable communication networks under such conditions. This study addresses this gap by proposing a hybrid communication system integrating Bluetooth Low Energy (BLE) with LoRa and LoRaWAN technologies. Our approach involves custom-developed hardware: LifeLinks with Renesas RA4W1 microcontrollers and LoRa-Router-Gateways equipped for both peer-to-peer and cloud connectivity. To validate our system, we conducted a series of experiments for high-traffic scenarios, network disruptions, diverse environmental conditions, and coordinated rescue operations. Results demonstrated 99% successful message delivery, minimal data loss, and strong network reliability, indicating the system's efficacy in disaster scenarios. Our research bridges the gap in current literature by offering a comprehensive, resilient communication solution that ensures continuous data flow and robust performance even in severely disrupted environments.

Show Less

A Crash Response System using LoRa-based V2X Communications

Bruno Carneiro; Daniel Macêdo Batista; Roberto Hirata Junior; Kifayat Ullah **All Authors**

Publisher: IEEE | 2023 | Conference Paper

AI Summary

- Crash response system uses LoRa-based vehicular communications to reduce road accident response times.
- System broadcasts beacons with vital info (accident type, location, people involved, speed) to emergency services.
- Simulation results show network can transmit beacons across several km with minimal latency and varying node density affects PDR and latency.

Abstract

Emergency services response time is a critical factor for road traffic injury care, and the current scenario is that most accidents must be reported by a bystander or by those involved, which usually takes several minutes. This work proposes a crash response system using LoRa-based vehicular communications, aiming to reduce response times for road accidents, save lives, and take advantage of the license-free radio frequencies used by LoRa. The system uses broadcast beacons to build multi-hop routes from vehicle nodes to emergency services. The beacon message contains important information, such as the type of accident, latitude, longitude, the estimated number of people, and the last known speed. We have evaluated our proposal in the ns3 network simulator by simulating an urban environment in a 10km wide area and varying the node density. Results have shown that the network can transmit a beacon across several kilometers with minimal latency. The amount of nodes in the network is a critical factor, with greater node density network instances displaying a larger Packet Delivery Ratio (PDR) and Network Coverage but also a slightly longer latency. As a complementary contribution, all the code produced in this research is publicly available as Open Source Software. Show Less

A LoRa based Emergency Communication Device for Disaster Response

Pradeep Kumar R; Diliban Selvarathinam K P; C.N. Gnanaprakasam **All Authors**

Publisher: IEEE | 2025 | Conference Paper

AI Summary

- Self-powered LoRa communication system for disaster scenarios uses Arduino hardware and hand-cranked generator.
- Device features GPS, oxygen sensor, and OLED display for real-time position and environmental info.
- Affordable, dependable, and energy-efficient solution enhances coordination in rescue operations with uninterrupted data transmission.

Abstract

During disaster scenarios, rescue efforts are delayed by communication breakdowns caused by inaccessibility of cellular networks and Wi-Fi. The proposed model overcomes the problem with a self-powered long-distance communication system utilizing LoRa technology and Arduino hardware. With a hand-cranked power generator, it operates continuously without needing an external power source. The device features a NEO-6M GPS chip and oxygen sensor to relay real-time position and environmental information over a LoRaWAN network. The proposed model has pre-formatted emergency messages appearing on an OLED display, which are relayed with sensor readings to a connected LoRa receiver via a Wi-Fi-enabled web interface and app. This system provides an affordable, dependable, and energy-efficient disaster management solution that facilitates real-time tracking and enhanced coordination in rescue operations. The study proves that the proposed model's internal power source and uninterrupted data transmission greatly improve situational awareness and response effectiveness in emergency situations. Show Less

Do IoT LoRa Networks Support Emergency Evacuation Systems ?

Aghiles Djoudi; Rafik Zitouni; Nawel Zangar; Laurent George **All Authors**

Publisher: IEEE | 2019 | Conference Paper

AI Summary

- Building sites need evacuation systems to quickly evacuate workers, as lack of mechanisms puts citizens' lives at risk.
- A novel IoT-based emergency application with LPWAN can address this issue, using LoRa network and 6LowPAN for mesh communication.
- The goal is to deploy an Emergency Evacuation System (EES) in building sites to improve worker safety.

Abstract

The number of building sites increased considerably these few last years. Emerging countries invest more and more in building and construction sites to overcome the need of citizens like education, residential and administration buildings. Such sites should be equipped with evacuation systems to allow bricklayers and other persons to be evacuated quickly. The lack of such mechanisms makes the life of billion of citizens in a danger. This problem could be addressed by building a novel evacuation emergency application in Internet of Things (IoT) devices with Low Power Wide Area Networks (LPWAN). One of the known LPWAN networks is Long Range (LoRa) network. To overcome the need of peer to peer communication between devices, IEEE.802.15.4 technology and particulary 6LowPAN could be used to allow devices to communicate in a mesh network. Our work is motivated by the deployment of an Emergency Evacuation Systems (EES) in building sites. Show Less

Enhancing Quality-of-Service in LoRa Low-Power Wide-Area Networks via Multi-Hop Optimized Radio Resource Management

Laith H. Jasim Alzubaidi; G Sunil; Hirald Dwaraka Praveena; N. Sathishkumar; Manjula B. M **All Authors**

Publisher: IEEE | 2024 | Conference Paper

AI Summary

- LoRa wireless communication optimizes transmission policy parameters for excellent QoS; current methods haven't fully optimized all parameters.
- MQ-LoRa uses a hybrid topology with cluster and tree structures, and a membrane-inspired clustering algorithm for efficient information sharing.
- The system uses SAC-based deep reinforcement learning to optimize transmission policies considering signal-to-interference ratio, QRank, and environmental factors.

Abstract

In transport systems in particular, LoRa wireless communication has become an essential technology for intelligent critical infrastructures. Optimizing transmission policy parameters, including carrier frequency, bandwidth, coding rate, and spreading factor, is crucial for achieving excellent Quality of Service (QoS) for LoRa systems; however, current methods have not fully optimized all LoRa parameters. In order to efficiently share information, the proposed Multiobjective Q Learning based LoRa (MQ-LoRa) uses a hybrid topology consisting of a cluster root rotating tree, wherein IoT nodes adhere to a cluster topology inspired by cellular tissues, while gateways follow a tree topology. For cluster construction, a membrane-inspired clustering algorithm is created, and influence scores are used to determine which header node is best. The study presents Data QoS Ranking for IoT nodes, which uses the new LoRa frame structure field QoS Ranking (QRank) to classify data into priority and non-priority categories. The Soft Actor Critic (SAC) fast deep reinforcement learning model is used for optimal transmission policy enforcement, taking into account signal-to-interference-plus-noise ratio, QRank, and other environmental characteristics. After that, the transmission policy is optimized taking carrier frequency, bandwidth, coding rate, and spreading factor into account. Furthermore, a multi-hop routing method with concurrent optimization that uses shuffling shepherd and mayfly optimization rates routes according to fitness standards. A multi-weighted sum approach is used to establish a weighted duty cycle in LoRa IoT networks to minimize resource inefficiency and loss of information. Using the NS3.26 LoRaWAN module, performance evaluation measures energy consumption, latency, throughput, packet receipt ratio, and packet rejection ratio, among other parameters. Comparison with popular LoRa techniques shows that the suggested MQ-LoRa performs better energy consumption of 56 mJ than exisitng Reinforcement Learning based LoRA, according to experimental results. Show Less

Poster Abstract: Evaluation of a LoRa Mesh Wireless Networking System Supporting Time-Critical Transmission and Data Lost Recovery

Chi-Wen Liang; Yung-Lin Wu; Cheng-Yu Shi; Shu-Min Lu; Huang-Chen Lee **All Authors**

Publisher: IEEE | 2019 | Conference Paper

AI Summary

- LoRa mesh network evaluated for large-area sensor monitoring.
- Data recovery mechanism developed for periodic data communication.
- Average packet delivery ratio achieved: 99.99% for periodic reporting data.

Abstract

This paper evaluates a LoRa mesh networking system for large-area monitoring of sensor applications. A prior study [1] proposed a LoRa-based wireless mesh network that supported periodic data collection. In this study, we further evaluated how data recovery mechanisms and time-critical data transmission affect periodic data communication, and a data transmission recovery mechanism was developed to recover lost data. Additionally, in emergency situations (e.g., detecting fire or alarm events), the sensor's data must be immediately sent to the control center. In a LoRa mesh network, these time-critical data packets, called non-periodic packets (NPPs), are designed to minimize transmission delays. We deployed 20 nodes on our university campus to form a LoRa mesh network, and the system achieved an average 99.99% packet delivery ratio with the data recovery mechanism for periodic reporting data. For NPPs, the average transmission delays achieved were 2.284 seconds, 4.597 seconds, and 6.75 seconds to deliver data to the gateway in one-hop, two-hop, and three-hop scenarios, respectively. Show Less

LoRa Based Wireless Network for Disaster Rescue Operations

Aishwarya Vithayathil; Afaf Mohamed; Ann Jacob; Chinju Merin Raju; Jaison Jacob **All Authors**

Publisher: IEEE | 2021 | Conference Paper

AI Summary

- Proposes Long Range (LoRa) based wireless network for disaster rescue operations.
- Provides offline communication system for victims to communicate with rescue teams.
- Uses LoRa and Wi-Fi technology for communication, enabling portable nodes and victim connectivity.

Abstract

Communication is very crucial during a crisis. It plays a vital role in rescuing victims at times of disasters. Catastrophic disasters can collapse conventional communication systems partially or completely, making rescue operations extremely difficult. Hence this requires an easily deployable, easily maintainable, independent communication system to help victims communicate to rescue teams. This paper presents a proposal of Long Range (LoRa) based wireless network for disaster rescue operations, to provide an offline communication system that gives the first responders a simple interface for managing the aspects of a disaster. The system uses LoRa and Wi-Fi technology for communication. LoRa is a wireless technology that offers low power, long-range and secure data transmission for machine-to-machine (M2M) and Internet of Things (IoT) applications. Portable wireless nodes can be installed in the disaster area and survivors can connect to these nodes using Wi-Fi devices and communicate their needs. The victim's messages are communicated to the rescue team using LoRa technology. A prototype system is constructed to evaluate the function of the network and its performance.

Show Less

The Design and Development of LoRa-LoRaWAN Wireless Network Infrastructure with Multi-Hop Capability

Yoga Prastiya Wibawa; Xerandy; MohAlma Samudro; Bondan Suwandi; Fito Wigunanto Herminawan; I Putu Ananta Yogiswara; Dimas Biwas Putr... **All Authors**

Publisher: IEEE | 2024 | Conference Paper

AI Summary

- Wireless mesh networks support multi-hop communication for reliable data delivery and broader coverage at low cost.
- LoRa is not designed for multi-hop capabilities by default, but a multi-hop LoRa technology is needed in areas with obstacles and remote areas.
- A multi-hop LoRa protocol integrated with LoRaWAN increases data transmission success rate up to 94.0%.

Abstract

Typically, a wireless mesh network supports multi- hop communication, enhancing data delivery reliability and broader area coverage at a low cost. LoRa, a digital and low- speed communication medium, is not designed to perform these capabilities by default. However, in some conditions, a multi- hop LoRa technology is needed, especially for application in areas with many obstacles and remote areas. To address this issue, this paper discusses designing and implementing a multihop LoRa protocol integrated with LoRaWAN. The multi-hop system increases the success rate of data transmission from the farthest node to the gateway up to 94.02%, which cannot be achieved without multi-hop. In the multi-hop setup, repeater nodes relay the LoRa device signals to subsequent LoRa devices until reaching the gateway. The tests demonstrate how messages flow from the transmitter node to the gateway, which is then forwarded to the LoRaWAN network services without requiring hardware modifications. The tests were conducted indoors under conditions with and without obstacles. Show Less

Emergency Communication in IoT Scenarios by Means of a Transparent LoRaWAN Enhancement

Emiliano Sisinni; Dhiego Fernandes Carvalho; Paolo Ferrari **All Authors**

IEEE Internet of Things Journal
Publisher: IEEE | 2020 | Journal

AI Summary

- LoRa-REP access method reduces average transaction time and increases success probability in emergency IoT scenarios.
- Two operational paradigms tested: public LoRaWAN infrastructure with cloud-based backend, and private LoRaWAN networks with local backend (edge/fog computing).
- Experimental results show increased success probability up to 99.5% and reduced average transaction time by up to 15%.

Abstract

This work deals with the management of sporadic and rare events linked with emergency situations in wireless Internet-of-Things (IoT) scenarios. The goal is to increase the performance of the emergency communication, when low-power wide area networks (LPWANs) are used as IoT backbone. In the proposed approach, a device usually operates as a normal node but, in case of emergency, can use the novel LoRa-REP access method. In this work, the LoRa-REP, based on message replication, is discussed focusing on its capability of reducing average transaction time and increasing success probability. Two operational paradigms have been considered and tested: public LoRaWAN infrastructure with cloud-based backend, and private LoRaWAN networks with local backend (edge/fog computing). Typical examples of public networks are smart cities, whereas local networks are often used in industry or building automation. Additionally, two real use cases (for public and local scenarios) are provided to show the effectiveness of the proposed approach. The experimental results (with a prototype device implementing LoRa-REP and named eNode) show that the success probability of the emergency communication can be increased up to 99.5%, and the average transaction time can be reduced up to 15% with respect to LoRaWAN without retries or up to 50% with respect to LoRaWAN with retries. Show Less

Rapidly Deployable Satellite-Based Emergency Communications Infrastructure

Francis Kagai; Philip Branch; Jason But; Rebecca Allen; Mark Rice **All Authors**

IEEE Access
Publisher: IEEE | 2024 | Journal

AI Summary

- Current satellite communication infrastructure fails to provide a single device that meets low power and low-cost rapid deployment requirements.
- The high cost of satellite applications is associated with spectrum licensing and satellite infrastructural costs, making deployable systems prohibitively expensive.
- Opportunities exist for developing low-power, frequency-agile architectures that integrate satellite and terrestrial network infrastructure.

Abstract

Communication infrastructure is crucial during emergencies because it ensures reliable coordination and information dissemination for effective search and rescue missions. However, these systems rely on physical structures, making them vulnerable to failures caused by disasters like fires and flooding. To enhance resilience, redundant infrastructure is commonly practiced. For instance, redundant systems ensure continuous operation in space missions where repairs are not feasible. A major gap exists, however, as redundancy is not a viable solution for remote areas without existing communication infrastructure. Additionally, victims may need to evacuate to safety zones not covered by mainstream communication systems. In such cases, alternative infrastructure that is easy to deploy is required to address emergency communication challenges fully. This paper presents a framework to analyze the deployability of current satellite communication infrastructure for this purpose. The analysis focuses on deployability dynamics, elements, and applications. We start by discussing the evolution of satellite ground stations into mobile and software-defined systems. In the analysis, we find that, despite these advancements, current infrastructure fails to provide a single device that meets low power and low-cost rapid deployment requirements based on our deployability framework. The product that almost fulfils this is the satellite phone. However, satellite phones are prohibitively expensive for large-scale deployments, with a single device costing over 600 United States Dollars (USD), more than ten times the cost of Internet of Things (IoT) devices supporting similar technologies. Additionally, they are not power efficient. The high cost of satellite applications is associated with spectrum licensing and satellite infrastructural costs. We have analyzed satellite communication frequency bands and architectural design, such as antenna designs, communication protocols and standards, modulation techniques, and signal processing methods, to understand the important factors toward full deployability. Our comprehensive analysis reveals a gap in addressing interoperability issues caused by disparate communication standards, processing power and energy requirements. Further analysis reveals promising open-source initiatives that offer potential solutions, such as low-data rate modulation schemes, low-bitrate voice codecs, and low-power encryption techniques. Additionally, advancements in Artificial Intelligence(AI) and robust cybersecurity techniques provide a solid foundation for future developments of rapidly deployable satellite-based emergency communication infrastructure. Consequently, we have identified opportunities for developing low-power, frequency-agile architectures that integrate satellite and terrestrial network infrastructure. Specifically, the next-generation Cospas-Sarsat Search and Rescue system provides a pathway for eliminating spectrum licensing costs, making deployable systems affordable for mass adoption. Future work should leverage advancements in low-bitrate technologies to enhance emergency systems, ensuring secure and resilient two-way voice, messaging, and distress signalling capabilities for critical communications. Show Less

Real-Time Disaster Response with LoRa: A Multi-Sensor Approach

Shubhi Agrawal; Dhruv Gupta; Vansh Sudan; Pratham Goel; Vaibhav Nijhawan **All Authors**

Publisher: IEEE | 2024 | Conference Paper

AI Summary

- Improves efficacy & efficiency during crises with LoRa based disaster response & management system.
- Enhances coordination & effectiveness of emergency services through sensors & GPS tracking solutions.
- Durable, adaptable, and resilient communication network in disaster-stricken areas using LoRa technology.

Abstract

The LoRa based disaster response and management system aims to create a strong and flexible communication network for emergency - responding teams. The primary goal is to improve efficacy and efficiency during crises. The implementation of LoRa technology represents an innovative solution to enhance the coordination and effectiveness of emergency services. The system facilitates quick deployment and enables continuous environmental surveillance via sensors and GPS tracking solutions. We have proposed an alarm system to share information with affected people and immediate respondents. LoRa (Long Range) technology is ideal for creating a durable, adaptable, and resilient communication network in disaster-stricken areas where traditional communication infrastructure may face risks or vulnerability. Overall, system helps to strengthen disaster preparedness and response mechanisms. Show Less

SDR-Based Reliable and Resilient Wireless Network for Disaster Rescue Operations

Dereje Mechal Molla; Hakim Badis; Alemayehu Addisu Desta; Laurent George; Marion Berbineau **All Authors**

Publisher: IEEE | 2019 | Conference Paper

AI Summary

- Survivors' connectivity ensured by multiple RANs carried by WMHN combining FMNs & MANETs.
- Each peer node in WMHN acts as a multi-technology access point for GSM, WiFi, LoRa.
- Prioritized serving radio technologies based on traffic type (SMS, voice, video) to enhance QoS.

Abstract

A reliable and resilient communication between survivors and the Search and Rescue (SAR) team, after a disaster, is critically important for efficient rescue operations. Unfortunately, in many cases, large-scale disasters like hurricanes, earthquakes and floods can damage the infrastructure of the local terrestrial wired and wireless networks. Several solutions have been proposed to provide reliable and resilient survivors connectivity using GSM base stations deployed by Flying Mesh networks (FMNs). The problem of this solution is that full coverage of the disaster area is not guaranteed and traffic having high bandwidth requirement cannot be supported. In this paper, we propose to offer survivors and rescuers connectivity through multiple radio access technologies (RANs). These RANs are carried by a Wireless Multi-Hop Network (WMHN) combining the existing FMNs with additional opportunistic Mobile Ad-Hoc Networks (MANETs) created by rescuers. We add to each peer node (in WMHN) the capability to be a multi-technology access point. GSM, WiFi and LoRa technologies are selected due to their native support by smartphones, and implemented by a single embedded Software Defined Radio (SDR) platform. In order to enhance the end-to-end Quality of Service (QoS) for survivors/rescuers, we prioritize the serving radio technologies based on the traffic type (SMS, voice and video). We show through simulation results that our solution reduces latency, packet loss and increases the connectivity of survivors. Show Less

A LoRa Based Reliable and Low Power Vehicle to Everything (V2X) Communication Architecture

Khandaker Foysal Haque; Ahmed Abdelgawad; Venkata P. Yanambaka; Kumar Yelamarthi All Authors

Publisher: IEEE | 2020 | Conference Paper

AI Summary

- A LoRa-based V2X communication architecture is proposed for reliable, robust, and low-power communication.
- The architecture enables vehicles to communicate reliably with roadside infrastructures at speeds of 10-30 MPH.
- The system can transmit data packets every 27-53 meters, allowing smooth transitioning between infrastructure connections.

Abstract

The industrial development of the last few decades has prompt to increase in the number of vehicles multi-fold. With the increased number of vehicles on the road, safety has become one of the major concerns. Inter vehicular communication, specially Vehicle to Everything (V2X) communication can address these pressing issues including autonomous traffic systems and autonomous driving. Extensive research is going on to develop a reliable V2X communication architecture with different wireless technologies like Long Range (LoRa) communication, Zigbee, LTE, and 5G. The reliability and effectiveness of V2X communication greatly depend on communication architecture and the associated wireless technology. In this conquest, a LoRa based reliable, robust, and low power V2X communication architecture is proposed in this paper. The communication architecture is designed, implemented, and tested in a real world scenario to evaluate its reliability. Testing and analysis suggest a vehicle in the road can communicate reliably with roadside infrastructures at different speeds ranging from (10-30) Miles per Hour (MPH) with the proposed architecture. At 10 MPH, a vehicle sends one data packet of 40 bytes every 27 meters and at 30 MPH, it sends the same data packet every 53 meters with smooth transitioning from communicating with one infrastructure to another. Show Less

Optimization and Performance Analysis of LoRa-Based Multi-Hop Networks Techniques

Jay Singh; Chandra Prakash Patidar All Authors

Publisher: IEEE | 2024 | Conference Paper

AI Summary

- LoRa technology shows promise for large-scale IoT deployments with low-power, long-range communication.
- The research focuses on optimizing and analyzing multi-hop networks for scalability, reliability, and efficiency.
- Secure communication is ensured between nodes and gateways using Rivest-Shamir-Adleman encryption.

Abstract

The Internet of Things is made possible by wireless communication technologies, which are essential for applications ranging from industrial automation to smart cities. In large-scale Internet of Things deployments, Long Range (LoRA) technology has shown promise as a low-power, long-range communication solution. The optimization and performance analysis of LoRa-based multi-hop networks are presented in-depth in this research, with an emphasis on improving scalability, reliability, and efficiency for various Internet of Things applications. The network design consists of gateways acting as middlemen for packet relaying data and nodes outfitted with LoRa transceivers. By ensuring secure communication between nodes and gateways; Rivest, Shamir, Adleman encryption protects the integrity and privacy of data. Important factors including packet delivery ratio, latency, scalability, and robustness are assessed through thorough simulation and analysis. Show Less

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